

INTRODUCTION

To design and develop an IoT-based transformer health monitoring system using ESP32 that continuously monitors temperature, oil level, and gas concentration, and automatically controls cooling fans and protection relay while displaying real-time data on a web interface accessible from anywhere.



CAUSES OF TRANSFORMER DAMAGE



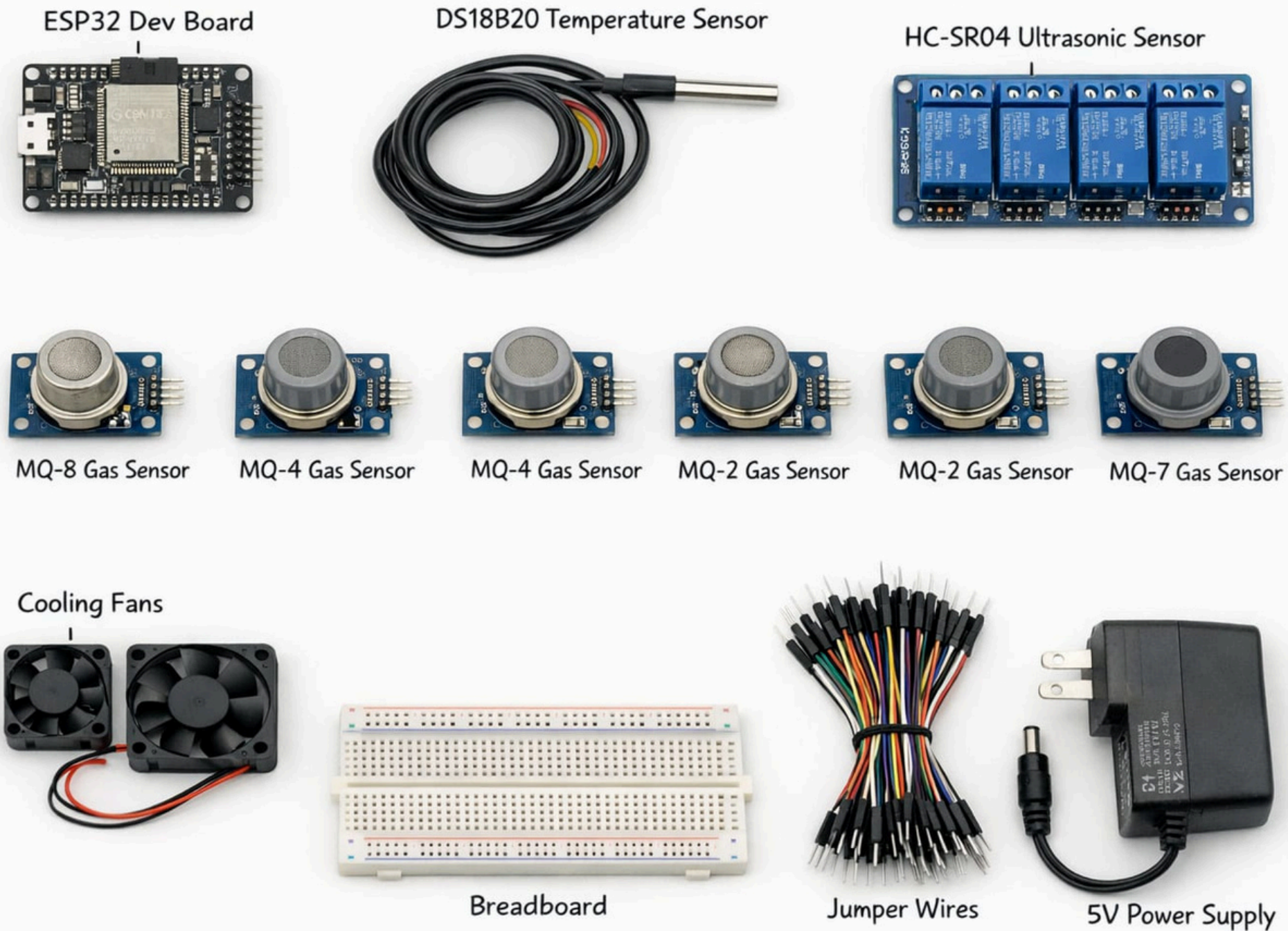
01 OVERHEATING

02 OIL LEVEL REDUCTION

03 GAS FORMATION DUE TO
INSULATION FAILURE



COMPONENTS REQUIRED



01 ESP32

02 RELAY MODULE

03 MQ-2, MQ-4, MQ-5, MQ-7, MQ-8 GAS
SENSORS

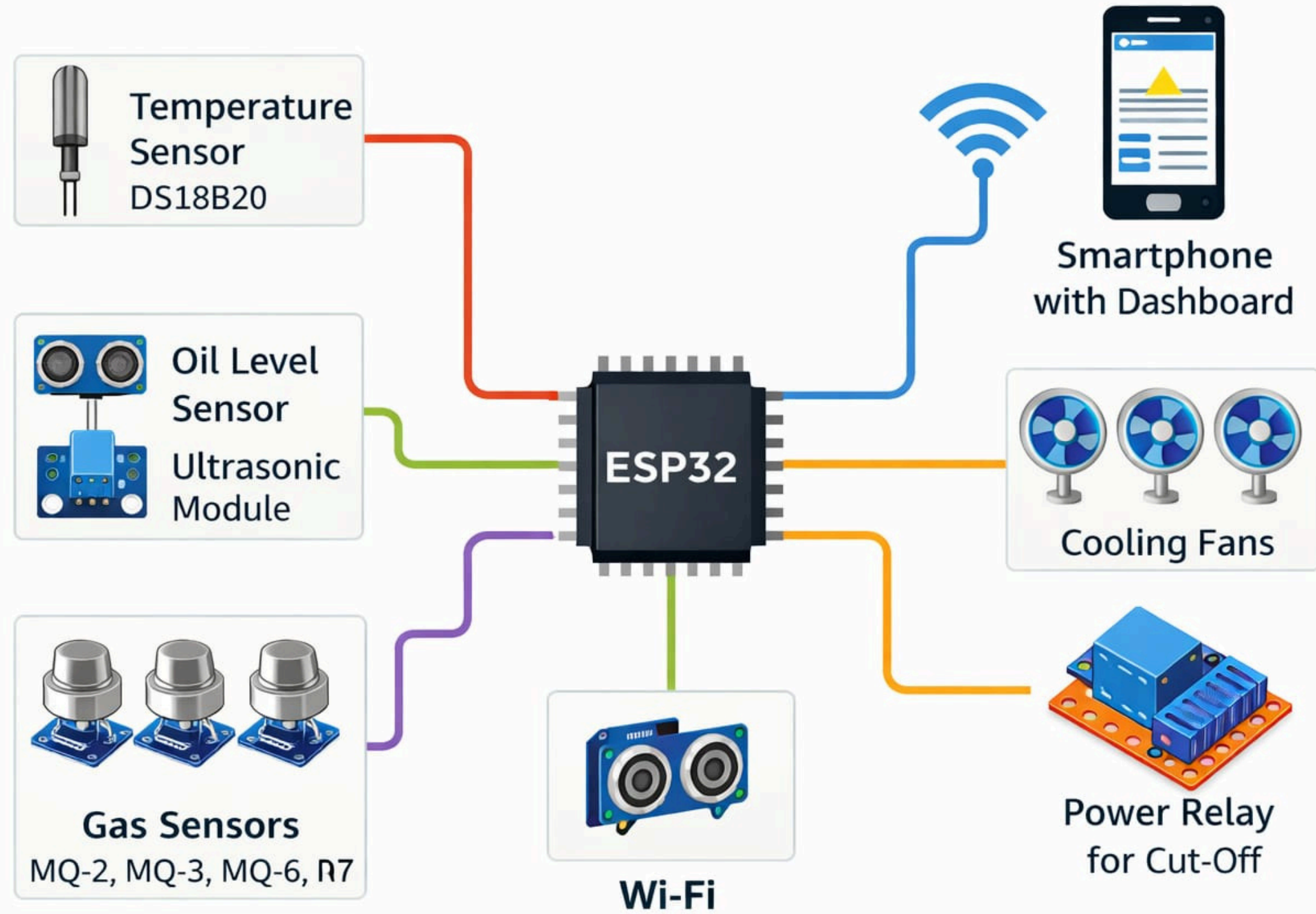
04 OIL LEVEL SENSOR

05 COOLING FANS (3)

06 TEMPERATURE SENSOR

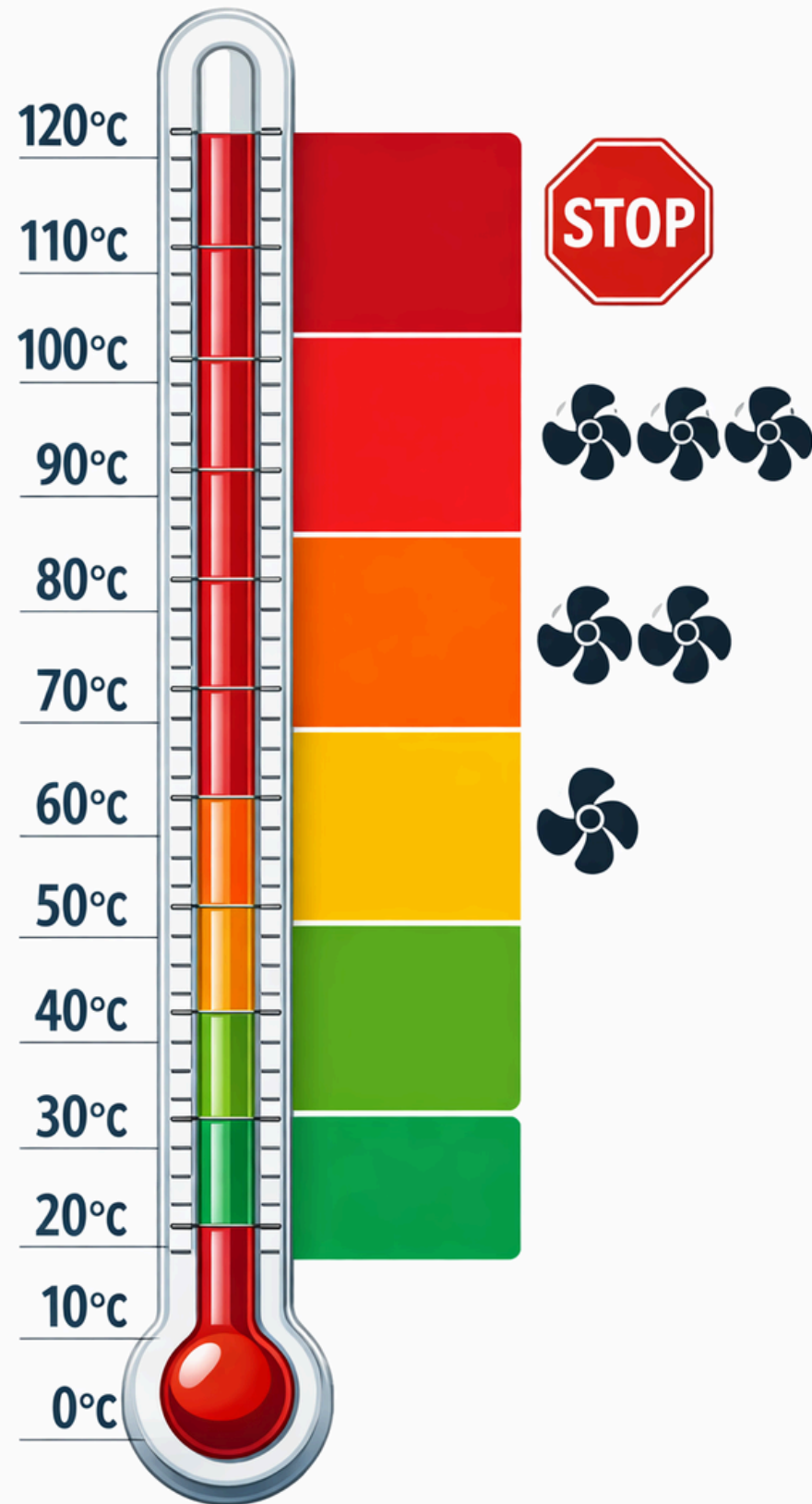
07 POWER SUPPLY

CIRCUIT DIAGRAM



IoT based  Transformer Monitoring System

TEMPERATURE MONITORING & COOLING

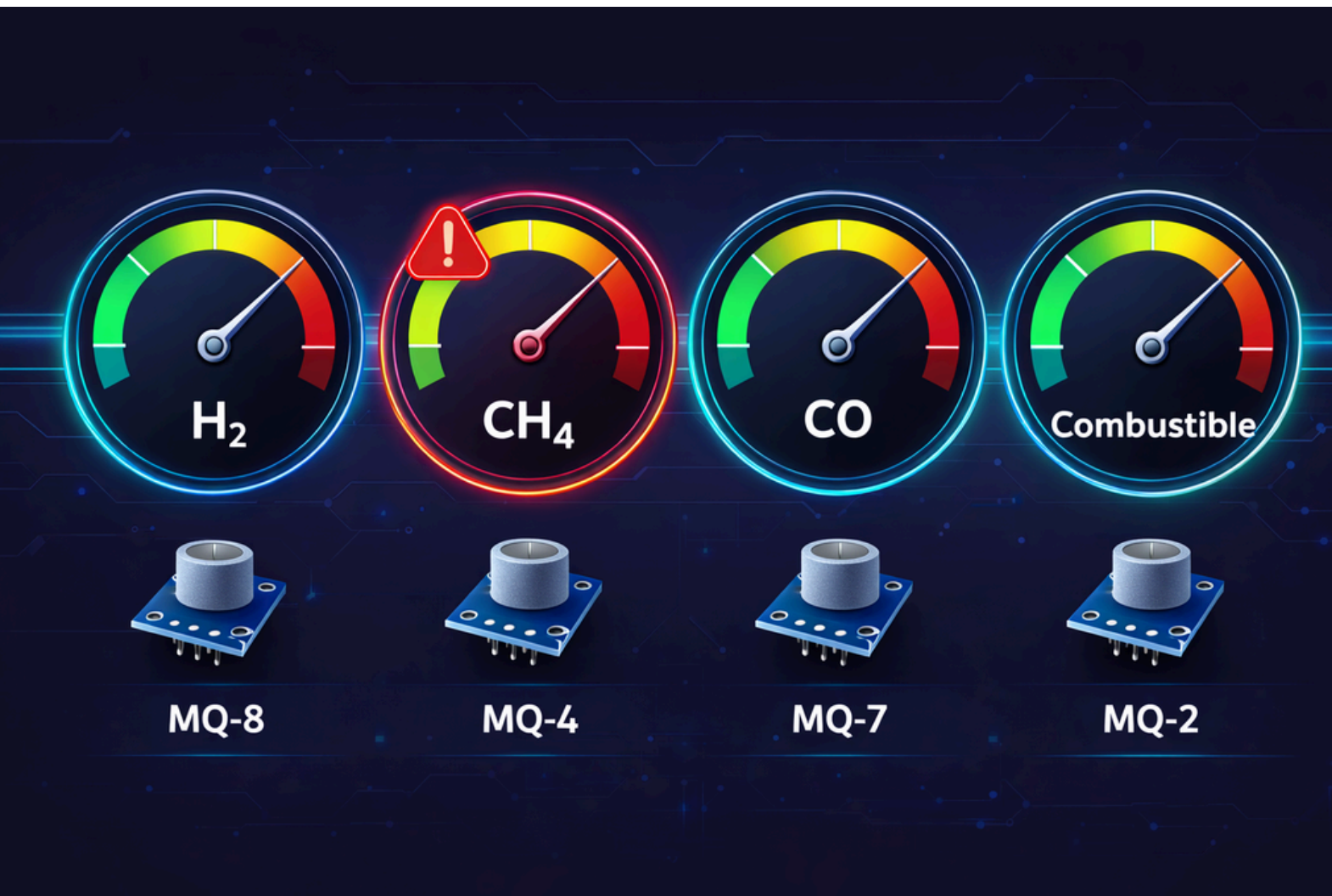


A temperature sensor (like DHT11/DS18B20) measures transformer temperature. Based on temperature:

- At 30°C → Fan 1 ON
- At 50°C → Fan 2 ON
- At 90°C → Fan 3 ON
- Above 100°C → Relay ON (cuts transformer supply)

This automatic cooling increases transformer life.

GAS DETECTION



Faults inside transformers produce gases like:

- Hydrogen (H₂) → detected by MQ-8
- Methane (CH₄) → detected by MQ-4
- Smoke/flammable gases → detected by MQ-2/MQ-5
- Carbon monoxide → detected by MQ-7
-

If gas levels exceed normal limits:

- System generates alert on webpage
- Indicates possible internal fault

OIL LEVEL MONITORING



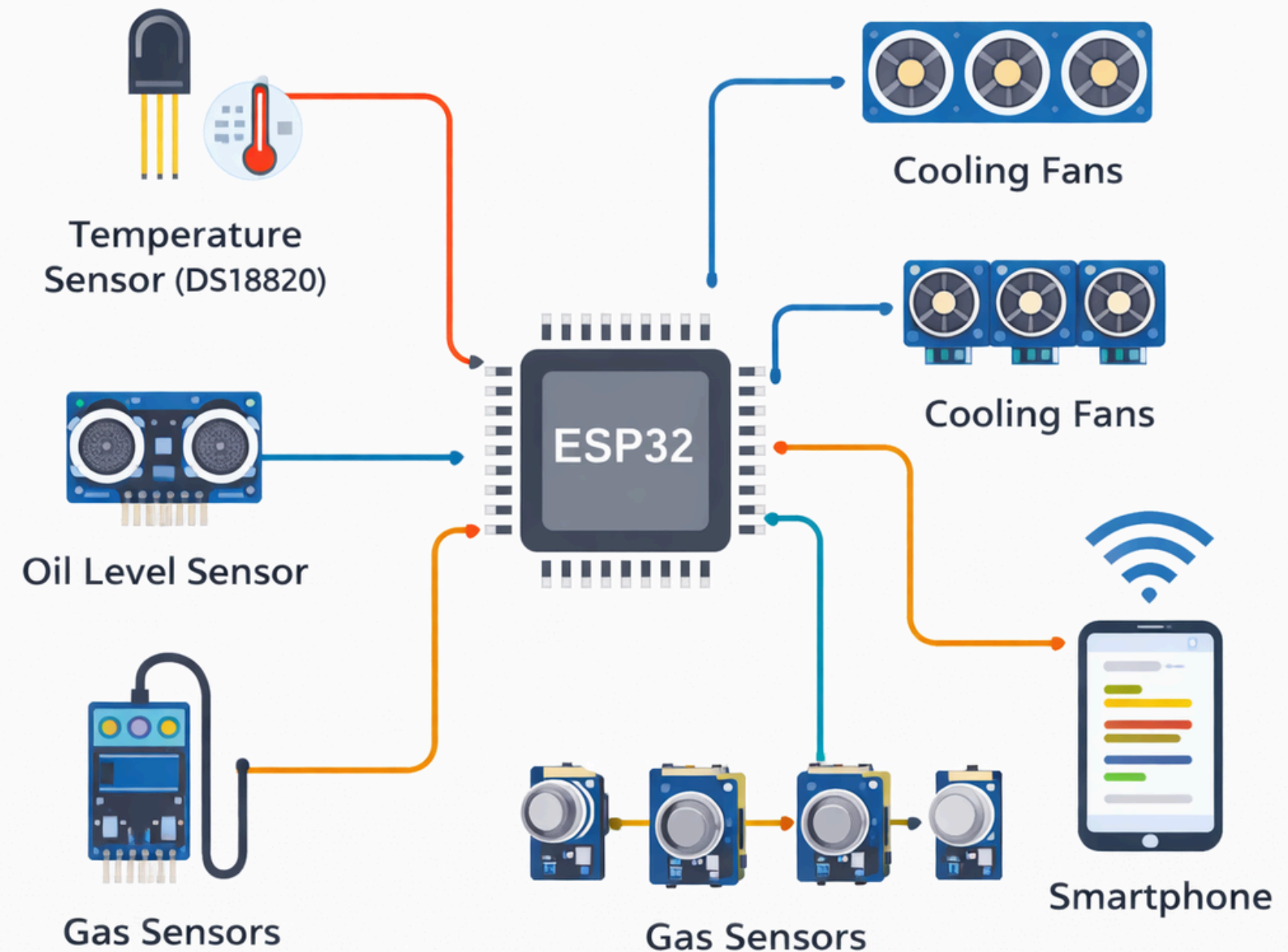
Transformer oil is essential for cooling and insulation.

- Oil level sensor detects oil quantity
- Low oil level → warning on webpage

IOT MONITORING USING ESP32

The ESP32 acts as the brain of the system:

- READS ALL SENSOR DATA
- CONTROLS RELAY AND FANS
- HOSTS A WEB SERVER
- SENDS DATA TO MOBILE VIA IP ADDRESS



FEATURES OF WEB DASHBOARD

- Live temperature display
- Fan status (ON/OFF)
- Gas levels (H₂, CH₄, CO, Smoke)
- Oil level indicator
- Alert messages (High Temp / Gas Leak)
- Relay status



ADVANTAGES

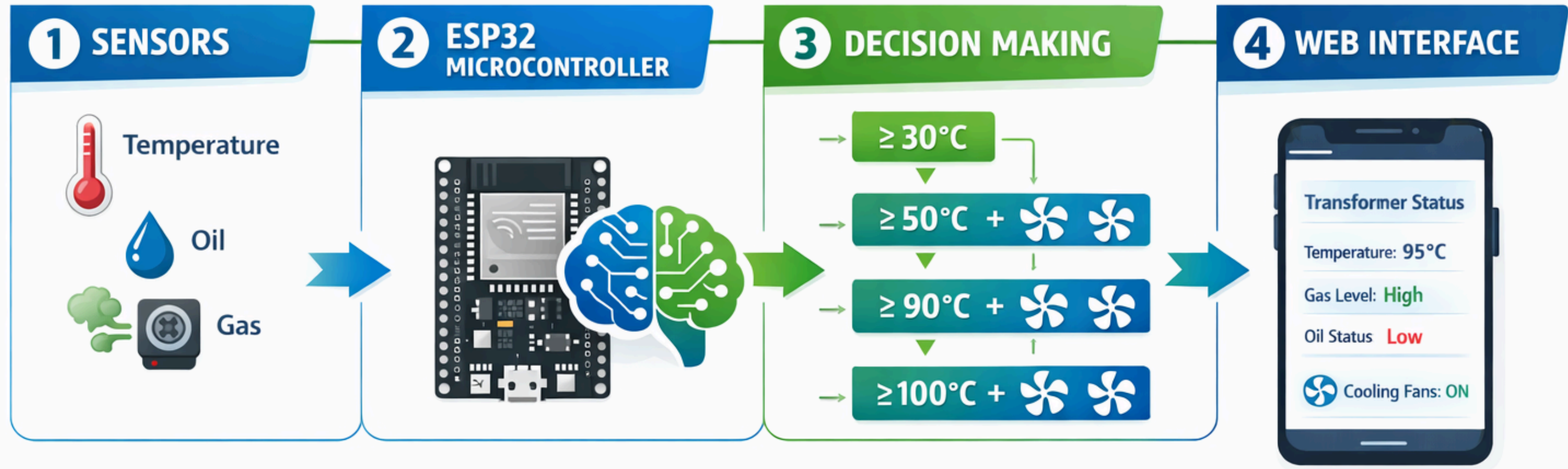
- Prevents transformer failure
- Real-time monitoring
- Remote access via mobile
- Automatic protection system
- Low-cost IoT solution

APPLICATIONS

- Power substations
- Industrial transformers
- Smart grid systems
- Renewable energy systems



AUTOMATIC TRANSFORMER MONITORING SYSTEM



THANK YOU

